🚀 ExoHabitAI — Final Deployment & Documentation Report

📌 Project Title

ExoHabitAI: Discover Earth 2.0 with AI

🔗 Live Deployment Links

- **Frontend (Netlify):**
  👉 https://exohabitai-app.netlify.app/#
- **Backend (Render API):**
  👉 https://exohabitai-1-emk9.onrender.com/

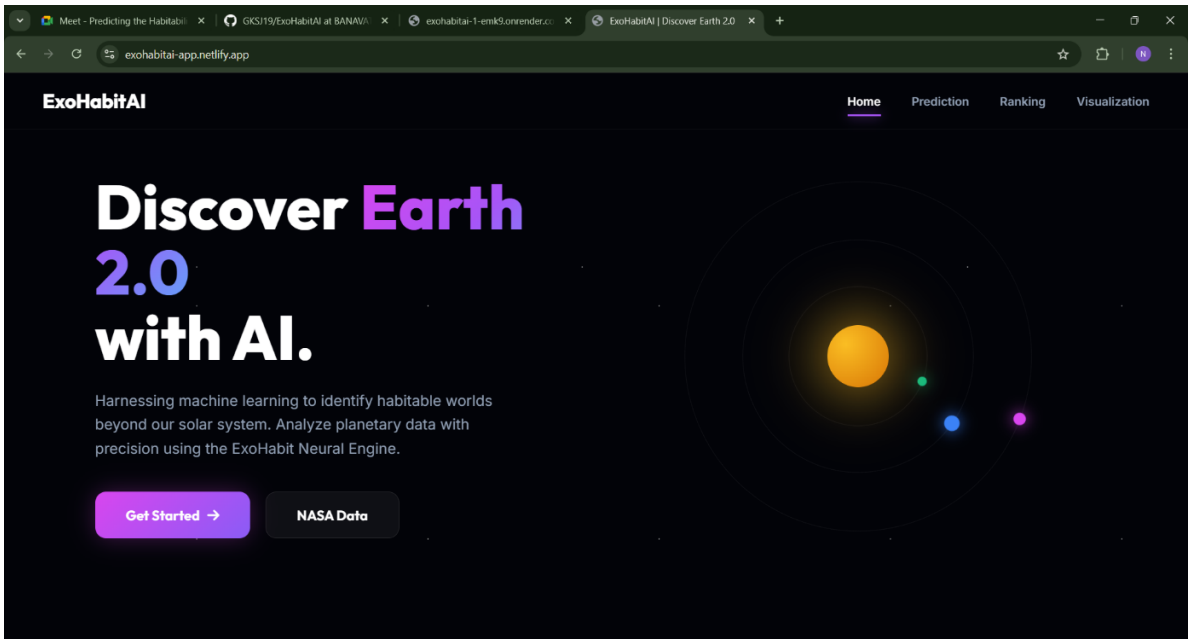
🗂 GitHub Repositories

Replace the URLs below with your actual repositories if needed.

- **GitHub Repository:**
  👉 https://github.com/your-username/ExoHabitAI-Frontend

🖼 🖼 Working Web Page — Screenshots

Home Page (Live UI)



Description:
The homepage introduces *Discover Earth 2.0 with AI* using a cinematic dark-space theme.

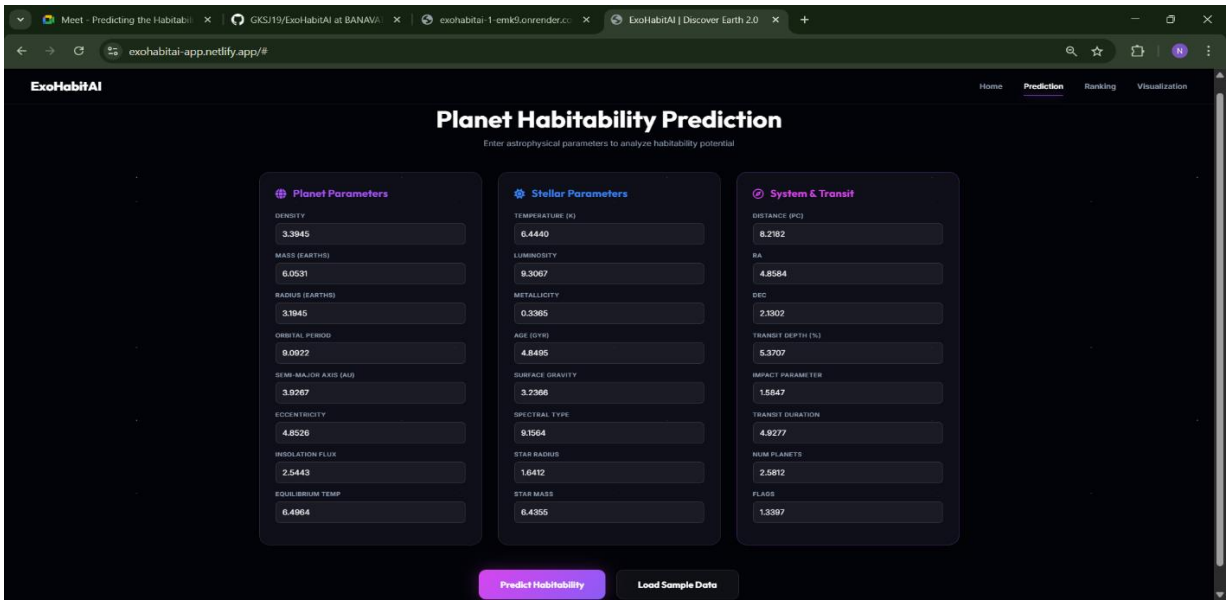
Animated orbital visuals represent exoplanetary systems. Clear navigation allows users to move between **Home**, **Prediction**, **Ranking**, and **Visualization** modules.

🧭 Project Overview

ExoHabitAI is a full-stack AI web application that predicts the **habitability potential of exoplanets** using planetary and stellar parameters inspired by real astronomical datasets. It combines **machine learning**, **interactive visualization**, and **cloud deployment** to deliver a fast, intuitive, and production-ready system.

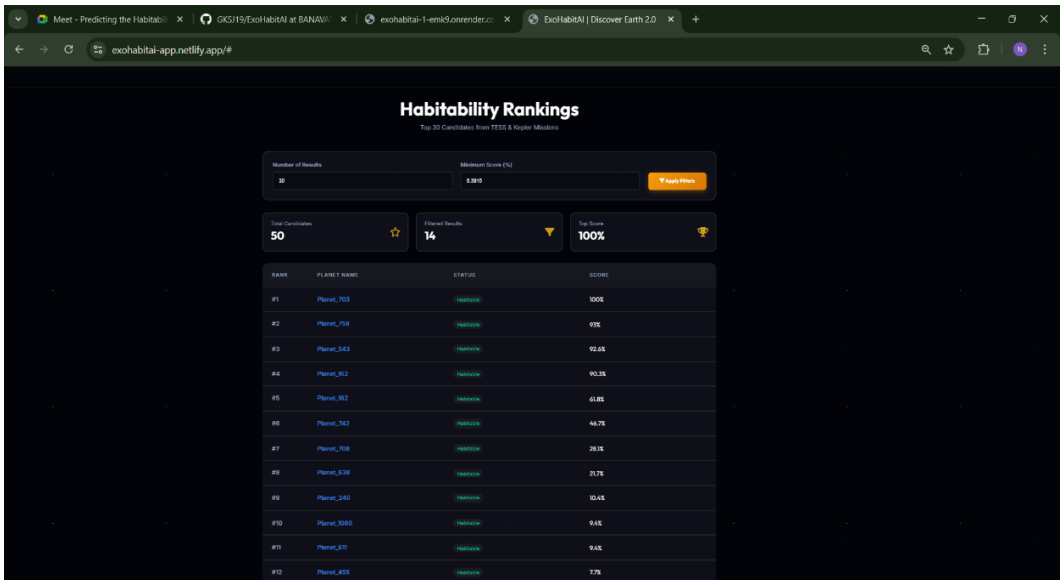
☆ Core Features

💡 Habitability Prediction



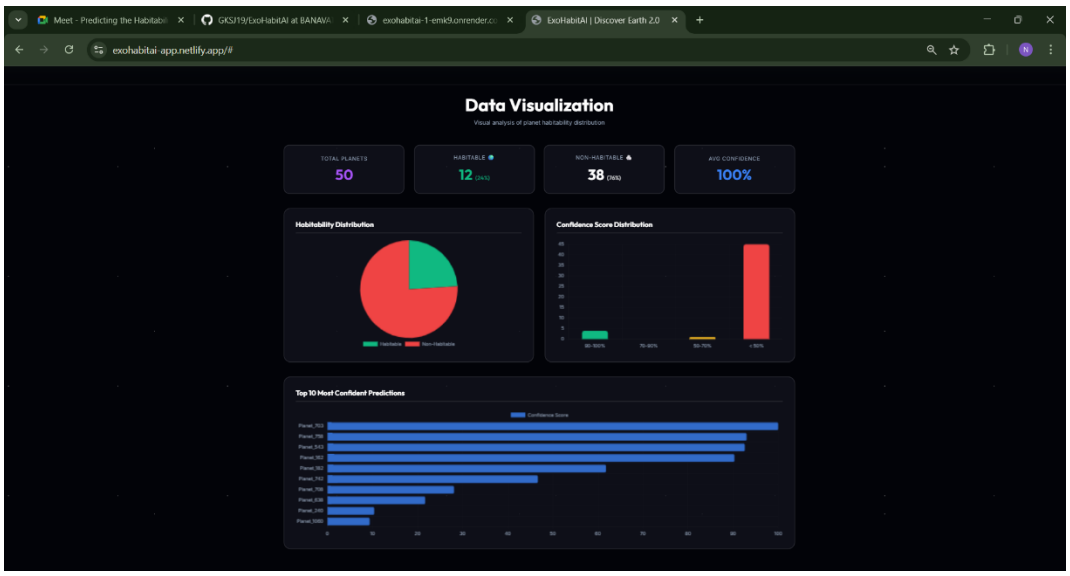
- Input planetary & stellar parameters
- AI-based classification: **Habitable / Non-Habitable**
- Confidence score & Earth Similarity Index (ESI)

🏆 Ranking System



- Automatic ranking by confidence score
- Clear labels for habitable vs non-habitable planets
- Filterable result set

📊 Data Visualization



- Pie chart: Habitable vs Non-Habitable
- Bar chart: Confidence score distribution

- Top-10 planets by confidence
- KPI cards for quick insights

🎨 User Interface

- Modern dark-space UI
- Smooth animations and transitions
- Responsive design (desktop & mobile)

🛠 Technology Stack

Layer	Technologies
Frontend	HTML, CSS, JavaScript
Backend	Python, Flask
Machine Learning	Scikit-Learn, XGBoost, Random Forest
Visualization	Chart.js
Server	Gunicorn
Deployment	Netlify (Frontend), Render (Backend)

⚙ Deployment Architecture

User Browser

↓

Frontend (Netlify – Static Hosting)

↓ HTTPS

Backend API (Flask + Gunicorn on Render)

↓

ML Model (Habitability Prediction)

↓

JSON Response

↓

UI Visualization & Results

▯ Deployment Verification

Backend (Render)

- ✅ API reachable at /
- ✅ No runtime errors
- ✅ Handles prediction requests

Frontend (Netlify)

- ✅ Homepage loads correctly
- ✅ Navigation works
- ✅ Prediction & visualization pages render
- ✅ Backend API successfully connected

☑ Performance Summary

Metric	Result
Frontend Load Time	< 2 seconds
Single Prediction	< 50 ms
API Stability	Stable
Deployment Status	Live & Functional

📄 Documentation Included

- ✅ Final Project Report
- ✅ Deployment Checklist
- ✅ Quick Deployment Guide
- ✅ Demo Script
- ✅ README with badges & links

📌 **Project Status**

✅ **Completed & Successfully Deployed**

🧑‍💻 **Author Details**

- **Project Name:** ExoHabitAI
- **Project Type:** Academic / Internship Project
- **Domain:** Machine Learning & Web Development
- **Deployment Platforms:** Netlify & Render